FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Acute Kidney Injury: AGN & Nephrotic Syndrome

Tutorial | Week 3 · Fits, TB, kidneys & nutrition | Paediatrics | Dr Aliku

Nephritic children (post-streptococcal GN) have cola urine, puffy eyes, hypertension and oliguria and need fluid and salt restriction, furosemide and BP control, while nephrotic children have massive proteinuria, low albumin and generalised oedema and need prednisolone; both can cause acute kidney injury, which you stage by creatinine and urine output, manage by getting the fluid balance right and fixing potassium, and refer for dialysis when AEIOU indications appear.

Must know before the ward round

1. Nephritic = Haematuria (cola urine), Hypertension, oedema (periorbital), Oliguria, mild proteinuria, raised creatinine. Nephrotic = proteinuria ≥3+ (UPCR ≥200 mg/mmol), albumin <30 g/L (IPNA/KDIGO; <25 g/L in older definitions), oedema, high cholesterol.
2. PSGN comes 1–2 weeks after GAS pharyngitis or 3–6 weeks after skin infection (impetigo, infected scabies). C3 is low and normalises within 8 weeks; ASO is raised after throat infection and anti-DNase B after skin infection.
3. PSGN management: restrict fluid to insensible losses (~400 mL/m²/day) plus urine output, restrict salt, give furosemide 1–2 mg/kg IV/oral, and treat BP (nifedipine/amlodipine; hydralazine IV). Penicillin for 10 days clears the strep but does not change the nephritis. No steroids.
4. Hypertensive encephalopathy in PSGN (headache, vomiting, blurred vision, seizures): lower BP by no more than 25% in the first 8 hours, give furosemide, and treat seizures.
5. Nephrotic initial episode (IPNA 2023 / KDIGO 2021 & 2025): prednisolone 60 mg/m²/day or 2 mg/kg/day (max 60 mg) once daily for 4 weeks, then 40 mg/m² or 1.5 mg/kg (max 40 mg) on alternate days for 4 weeks (8–12 weeks total).
6. Remission = dipstick negative/trace (UPCR <20 mg/mmol) for 3 consecutive days. Relapse = dipstick ≥3+ (UPCR ≥200 mg/mmol) for 3 consecutive days after remission. Steroid-resistant = no remission after 4 weeks of full-dose prednisolone.
7. Frequent relapses = ≥2 relapses in the first 6 months after initial response, or ≥4 in any 12 months. Steroid-dependent = 2 consecutive relapses during steroid therapy or within 14 days of stopping.
8. Nephrotic complications: infection (SBP, cellulitis: pneumococcus), thrombosis, hypovolaemia (abdominal pain, cold peripheries, high Hct), AKI, hyperlipidaemia and steroid toxicity. Screen for TB, HIV, hepatitis B and syphilis, and deworm before steroids.
9. KDIGO AKI staging: Stage 1 = creatinine 1.5–1.9× baseline or ≥26.5 µmol/L rise in 48 h, or UO <0.5 mL/kg/h for 6–12 h. Stage 2 = 2.0–2.9×, or UO <0.5 for ≥12 h. Stage 3 = ≥3×, ≥353.6 µmol/L, dialysis, eGFR <35 in under-18s, or UO <0.3 for ≥24 h or anuria ≥12 h. (NICE uses >8 h for children.)
10. Hyperkalaemia K^+ ≥6.5 or ECG changes: calcium gluconate 10% 0.5 mL/kg IV over 5–10 min (max 20 mL), salbutamol neb 2.5–5 mg, insulin 0.1 U/kg + glucose 0.5 g/kg, then remove K^+ (resonium, dialysis).
11. Dialysis indications (AEIOU): refractory Acidosis, Electrolytes (K^+ ≥7 or refractory), Intoxication, Overload (pulmonary oedema, >10–20% fluid overload), Uraemia (encephalopathy, pericarditis, bleeding), plus prolonged anuria. Peritoneal dialysis is the usual method for children in Uganda.
12. Bedside Schwartz eGFR (mL/min/1.73 m²) = 36.5 × height (cm) ÷ creatinine (µmol/L).

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In a nutshell

The kidney can fail a child in three common ways you will see on the Mengo/Mulago wards:

1. **Acute glomerulonephritis (AGN)**, usually **post-streptococcal (PSGN)**: the glomeruli are **inflamed**. Blood leaks into the urine (**cola-coloured**), salt and water are retained (**puffy eyes, hypertension, heart failure**), and urine output falls. It is the classic **nephritic syndrome**.
2. **Nephrotic syndrome**: the glomerular filter becomes **leaky to protein** without much inflammation. Massive **proteinuria → low albumin → generalised oedema**. In most children it is **minimal change disease** that melts away with **prednisolone**, but African children have more steroid-resistant and secondary disease.
3. **Acute kidney injury (AKI)**: a sudden fall in kidney function from any cause. In Uganda the commonest causes are **dehydration (diarrhoea)**, **severe malaria**, **sepsis**, **PSGN** and **nephrotoxins** (gentamicin, NSAIDs, herbs).

All three need the same core skills: **measuring urine output and weight, getting fluids right, controlling BP, and spotting dangerous potassium**.

> **CLINICAL PEARL**
> The one thing never to forget: **a puffy child is not always a "kidney child", and an oedematous kidney child is not always fluid-overloaded.** The nephritic child is **overfilled** (hypertensive, needs fluid restriction and furosemide). The nephrotic child may be **underfilled in the vessels** despite huge oedema (cold hands, tachycardia, abdominal pain), and furosemide alone can tip them into shock and AKI.

Definitions & classification

Nephritic vs nephrotic

Feature	Nephritic (e.g. PSGN)	Nephrotic (e.g. minimal change)
Typical age	5–12 years (UCG: >3 years)	1–6 years (minimal change)
Onset	Acute, after strep throat or skin infection	Insidious or after a viral illness
Urine	**Haematuria** (cola/tea-coloured), RBC casts, protein 1–2+	**Protein ≥3+**, frothy, little or no blood
Oedema	**Periorbital**, mild	**Generalised**, pitting, ascites, scrotal/labial, pleural effusions
BP	**High**	Usually normal (may be low if hypovolaemic)
Urine output	**Reduced**	Reduced (sodium retention)
Serum albumin	Normal or slightly low	**<30 g/L** (often <20)

Feature	Nephritic (e.g. PSGN)	Nephrotic (e.g. minimal change)
Cholesterol	Normal	**High**
Complement C3	**Low** (PSGN), normal by 8 weeks	Normal
Volume status	**Overloaded**	Often **intravascularly depleted**

Nephrotic syndrome definitions (IPNA 2023 / KDIGO 2021, 2025)

Term	Definition
Nephrotic-range proteinuria	First-morning **UPCR ≥200 mg/mmol (≥2 mg/mg)**, or ≥40 mg/m²/h (≥1 g/m²/day), or **dipstick ≥3+**
Nephrotic syndrome	Nephrotic-range proteinuria + **hypoalbuminaemia (<30 g/L)** or oedema when albumin is unavailable
Complete remission	UPCR **≤20 mg/mmol** or dipstick **negative/trace** on **3 consecutive days**
Steroid-sensitive (SSNS)	Complete remission within **4 weeks** of standard-dose prednisolone
Steroid-resistant (SRNS)	**No complete remission after 4 weeks** of standard prednisolone (IPNA uses a further confirmation period of up to 2 weeks before labelling SRNS)
Relapse	Dipstick **≥3+** (or UPCR ≥200 mg/mmol) on **3 consecutive days** after remission
Infrequent relapse	1 relapse within 6 months of initial response, or 1–3 relapses in any 12 months
Frequently relapsing (FRNS)	**≥2 relapses in the first 6 months** after initial response, or **≥4 relapses in any 12 months**
Steroid-dependent (SDNS)	**2 consecutive relapses during** steroid therapy (or alternate-day therapy), or **within 14 days of stopping**

AKI staging (KDIGO 2012, still the current AKI guideline)

Stage	Serum creatinine	Urine output
1	**1.5–1.9 × baseline** (within 7 days), or rise **≥26.5 µmol/L (0.3 mg/dL) within 48 h**	**<0.5 mL/kg/h for 6–12 h**
2	**2.0–2.9 × baseline**	<0.5 mL/kg/h for **≥12 h**
3	**≥3.0 × baseline**, or creatinine **≥353.6 µmol/L (4 mg/dL)**, or **start of dialysis**, or in under-18s **eGFR <35 mL/min/1.73 m²**	**<0.3 mL/kg/h for ≥24 h**, or **anuria ≥12 h**

- **pRIFLE** (paediatric RIFLE) uses a fall in estimated creatinine clearance (25% Risk, 50% Injury, 75% Failure) and urine output <0.5 mL/kg/h for 8 h (Risk), 16 h (Injury), or <0.3 for 24 h/anuria for 12 h (Failure).
- **NICE NG148** (AKI, updated 2019/2024) accepts pRIFLE, AKIN or KDIGO and defines a urine output criterion of <0.5 mL/kg/h for more than **8 hours in children and young people**.
- **Oliguria**: <0.5 mL/kg/h in children (<1 mL/kg/h in infants; UCG 2023 uses <1 mL/kg/h for all children). **Anuria**: no urine (UCG: <100 mL/24 h).
- **Bedside Schwartz eGFR** = **36.5 × height (cm) ÷ serum creatinine (µmol/L)** (or 0.413 × height ÷ creatinine in mg/dL).
- **Normal creatinine** rises with muscle mass: roughly 20–35 µmol/L in infants, 25–50 µmol/L at 1–5 years, 35–60 µmol/L at 5–12 years. "Normal for an adult" can mean stage 3 AKI in a toddler.

UGANDA CONTEXT

Most Ugandan children have **no baseline creatinine**. Use the **age-expected normal** (or back-calculate from an assumed eGFR of 100–120) and the **urine output** criterion. UCG 2023 calls AKI "acute renal failure" and defines oliguria as <1 mL/kg/h. Laboratory creatinine is available at most hospitals; potassium and blood gases mainly at RRH/NRH.

Epidemiology & Uganda context

- **PSGN** is common in Uganda because of **impetigo, infected scabies and untreated sore throats**, crowding and poor hygiene. It peaks in school-age children and is more common in boys. Skin-associated PSGN is particularly common in the tropics.
- **Nephrotic syndrome**: in Europe, ~85–90% of children are steroid-sensitive (mostly minimal change). In **sub-Saharan Africa** series, more children are **older at onset, steroid-resistant and have FSGS**, and **secondary causes** matter: **HIV-associated nephropathy, hepatitis B (membranous), quartan malaria (*P. malariae*), schistosomiasis (*S. mansoni*: MPGN), syphilis**. Steroid resistance in East African series is often reported around 20–40% (VERIFY local Mulago data).
- **AKI**: common causes are **dehydration from diarrhoea**, **severe malaria** (haemoglobinuria/"blackwater", hypovolaemia, sepsis; AKI is frequent in Ugandan children with severe malaria and predicts death), **sepsis**, **PSGN**, **nephrotoxins** (gentamicin, NSAIDs, herbal medicines), **sickle cell disease**, **tumour lysis**, and **obstruction** (posterior urethral valves in infant boys, stones, Burkitt lymphoma).
- **Services**: paediatric nephrology at **Mulago National Referral Hospital** with acute **peritoneal dialysis** (PD) for children; haemodialysis for older children at Mulago and a few other centres (VERIFY current paediatric capacity). Most regional hospitals cannot dialyse children, so **early referral** is essential.

Pathophysiology

PSGN (a type III hypersensitivity)

1. Nephritogenic GAS strains (pharynx or skin) release antigens (e.g. **nephritis-associated plasmin receptor, streptococcal pyrogenic exotoxin B**) that deposit in glomeruli.
2. Antibodies bind and form **immune complexes** (subepithelial "**humps**" on electron microscopy) → **complement activation (alternative pathway) → low C3**.
3. Inflammation and proliferation of glomerular cells → **reduced GFR → sodium and water retention** → expanded plasma volume → **oedema, hypertension, circulatory congestion** (pulmonary oedema, heart failure) and **hypertensive encephalopathy**.
4. Damaged capillaries leak red cells → **haematuria with dysmorphic RBCs and RBC casts**.
5. Latency: **1–2 weeks after pharyngitis**, **3–6 weeks after skin infection**. Unlike rheumatic fever, PSGN follows **skin** infection too.

Nephrotic syndrome

1. In minimal change disease, a circulating factor (T-cell/B-cell dysregulation, anti-nephrin antibodies in some) **effaces podocyte foot processes**, so the filter loses its charge/size barrier.
2. Massive **albuminuria → hypoalbuminaemia** → low oncotic pressure; plus **primary renal sodium retention** (the "overfill" component, via epithelial sodium channel activation).
3. Fluid moves into tissues → **oedema**. The intravascular volume may be **normal, high or low**: children with very low albumin and rapid onset can be **hypovolaemic** ("underfill").

4. The liver makes more proteins, including **lipoproteins** (→ hypercholesterolaemia) and **clotting factors**, while **antithrombin III** is lost in urine → **hypercoagulability** (renal vein thrombosis, cerebral venous sinus thrombosis).
5. Loss of **immunoglobulins and complement factors (factor B)** in urine, plus steroids → **infection** with encapsulated bacteria (spontaneous bacterial peritonitis, cellulitis).

AKI

Type	Mechanism	Ugandan examples	Clue
Pre-renal	Low kidney perfusion; kidney intact	Diarrhoea/dehydration, shock, sepsis, hypovolaemic nephrotic, heart failure	Concentrated urine, **urine Na <20 mmol/L**, **FENa <1%**, responds to fluid
Intrinsic renal	Damage to tubules, glomeruli, interstitium	**Acute tubular necrosis** (prolonged pre-renal, sepsis, malaria, haemoglobinuria), **PSGN**, nephrotoxins, HUS, TLS	Urine Na >40, FENa >2%, casts, blood/protein
Post-renal	Obstruction	Posterior urethral valves, stones, bladder outlet, tumour	Palpable bladder, hydronephrosis on ultrasound

Fractional excretion of sodium (FENa) = (urine Na × plasma creatinine) ÷ (plasma Na × urine creatinine) × 100. It is unreliable after furosemide.

Clinical acumen

Focused history

Ask about	Why it matters
Urine colour (cola, tea, red), **volume** (nappies, last void), frothiness	Haematuria vs haemoglobinuria (malaria, G6PD); oliguria; proteinuria
Sore throat 1–2 weeks ago, skin sores/scabies 3–6 weeks ago	PSGN
Swelling: where it started (eyes in the morning), speed, abdomen, scrotum	Nephrotic vs nephritic; ascites
Headache, vomiting, blurred vision, fits	**Hypertensive encephalopathy**
Breathlessness, cough, orthopnoea	Pulmonary oedema, effusions, heart failure
Abdominal pain in a nephrotic child	**Peritonitis** (SBP) or **hypovolaemia**, thrombosis
Diarrhoea, vomiting, fever, malaria, sepsis	Pre-renal AKI; malaria AKI
Drugs: gentamicin, NSAIDs, herbal medicines, ACE inhibitors	Nephrotoxic AKI
Rash, joint pain, abdominal pain, bloody diarrhoea	HSP (IgA vasculitis), SLE, HUS
Previous episodes, steroids (dose, response, side effects)	Relapse vs new; steroid dependence
HIV status (child, mother), hepatitis B, TB contact, schistosomiasis (lake swimming)	Secondary nephrotic syndrome; TB must be treated before high-dose steroids
Poor urine stream in an infant boy, dribbling, UTIs	Posterior urethral valves
Growth, feeding, immunisation (PCV, measles, varicella)	Malnutrition (kwashiorkor oedema mimic), infection risk on steroids

Examination

- **Vitals**: BP **with the right cuff** (bladder width ~40% of arm circumference) and compare with **age/height centiles** (AAP 2017: hypertension ≥95th centile; from 13 years ≥130/80), HR, RR, SpO_2, temperature, **CRT**, **weight (compare with the last known weight)**.
- **Volume status**: JVP (older child), liver size, gallop, basal crackles, oedema (periorbital, sacral, legs, scrotum/labia, **ascites** with shifting dullness, **pleural effusions**), and **peripheral perfusion** (cold hands + tachycardia + oedema = underfilled nephrotic).
- **Skin**: **impetigo scars, scabies**, purpura (HSP), malar rash (SLE), pallor (HUS, CKD, malaria).
- **Abdomen**: tenderness/guarding (peritonitis), renal masses, **palpable bladder** (obstruction).
- **Neuro**: conscious level, fundoscopy (hypertensive retinopathy), focal signs (thrombosis, PRES).
- **Growth**: weight, height, MUAC (oedema falsely raises weight and MUAC), Cushingoid features on long-term steroids.

Red flags

RED FLAG

- **Hypertensive emergency**: headache, vomiting, visual loss, seizures, reduced consciousness, BP far above the 99th centile.
- **Pulmonary oedema**: orthopnoea, crackles, SpO_2 <90%, gallop.
- **Hyperkalaemia**: K^+ ≥6.5 mmol/L or ECG changes (peaked T, wide QRS, arrhythmia).
- **Anuria**, rising creatinine, uraemic encephalopathy, pericardial rub.
- **Nephrotic child with abdominal pain + fever** (peritonitis), or **cold peripheries, tachycardia, oliguria** (hypovolaemia), or **sudden headache/seizure/limb swelling** (thrombosis).
- **Rapidly progressive GN**: creatinine doubling over days, persistent oliguria (crescents): urgent nephrology referral.

Thinking like a clinician

- **Cola urine + puffy eyes + high BP after a skin infection** = PSGN. Check C3 and anti-DNase B. If the C3 is normal, or the child does not improve within 2 weeks, think **IgA nephropathy, HSP, MPGN, lupus, or RPGN**.
- **Massive oedema in a 3-year-old with 4+ protein, no blood, normal BP** = minimal-change-type nephrotic syndrome. If **older (>12 years) or under 1 year, persistent hypertension, gross haematuria, low C3, raised creatinine, or HIV/HBV positive** = **atypical**: think biopsy and referral before (or instead of) steroids.
- **Oedema without proteinuria**: think **kwashiorkor**, heart failure, liver disease, protein-losing enteropathy.
- **Red urine + dipstick blood ++ but no RBCs on microscopy** = **haemoglobinuria** (malaria, G6PD haemolysis, blackwater) or **myoglobinuria**.
- **Nephrotic + furosemide + abdominal pain + rising Hct/creatinine** = **hypovolaemia**. Stop the diuretic; give albumin if available.
- **Oliguria in a dehydrated child**: give fluid first and reassess; "oliguria" that responds to rehydration was pre-renal.
- **Common student mistakes**: (1) giving maintenance IV fluids to an oliguric PSGN child; (2) giving furosemide to a shocked nephrotic child; (3) prescribing gentamicin at full dose in AKI; (4) forgetting to screen for TB before months of prednisolone.

Clinical workup

Step 0: First 5 minutes

1. **ABC**: SpO_2, work of breathing. **Pulmonary oedema** → sit up, oxygen, **furosemide 1–2 mg/kg IV**.
2. **Seizure or encephalopathy** → treat the seizure (diazepam 0.3 mg/kg IV or 0.5 mg/kg rectal; or midazolam), **check glucose**, **measure BP** (hypertensive encephalopathy) and **sodium** (hyponatraemia).
3. **Circulation**: shock (dehydration, sepsis, underfilled nephrotic) → **10–20 mL/kg** Ringer's/NS bolus and reassess (in a nephrotic child, **albumin 20% 0.5–1 g/kg** if available). **Do not bolus** an overloaded hypertensive nephritic child.
4. **Potassium**: if K^+ ≥6.5 mmol/L or ECG changes, start hyperkalaemia treatment now.
5. **Urine**: dipstick, catheterise if needed to measure output, start an **input/output chart** and **daily (or twice-daily) weights**.
6. **Bloods**: creatinine/urea, Na, K, glucose, Hb, malaria RDT.

Step 1: Focused history

- Infections (throat 1–2 weeks, skin 3–6 weeks), urine changes, swelling, headache/vision/seizures, breathing.
- Diarrhoea, vomiting, fluid intake; drugs and herbs.
- Previous episodes, steroid course, relapses.
- HIV, TB, hepatitis B, malaria, schistosomiasis exposure.
- Immunisation (PCV, measles, varicella), feeding, growth.

Step 2: Focused examination

- Weight (compare), height, BP centile, HR, RR, SpO_2, CRT, temperature.
- Oedema map, ascites, effusions, JVP, liver, gallop, crackles.
- Skin infections, rash, purpura.
- Abdominal tenderness (peritonitis), bladder, masses.
- Fundi, neurology.
- Document: "Periorbital oedema, BP 138/92 (>99th centile for height), no crackles, liver 2 cm, healed impetigo on both legs, urine output 0.4 mL/kg/h over 8 h."

Step 3: Bedside tests

Test	What it shows
Urine dipstick	Blood + protein 1–2+ (nephritic); protein 3–4+ with little blood (nephrotic); SG (concentrated in pre-renal); leucocytes/nitrites (UTI)
Urine microscopy (if a microscope is available)	**RBC casts** and dysmorphic RBCs = glomerular bleeding; no RBCs with dipstick blood = haemoglobin/myoglobin
Urine output (catheter, nappy weights: 1 g = 1 mL)	Oliguria staging
Weight (same scale, same time)	Fluid balance; best single guide to overload
BP with correct cuff	Hypertension
HemoCue Hb, RBS, malaria RDT	Anaemia, hypoglycaemia, malaria
ECG (if K^+ high or unknown with oliguria)	Peaked T waves, wide QRS, arrhythmia
Point-of-care ultrasound	Bladder volume, hydronephrosis, ascites, pleural effusion

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management
Urea, creatinine, Na, K	All three conditions; daily while unstable (6–12-hourly if K high)	Raised creatinine (AKI); **K^+ ≥6.0**; hyponatraemia (dilutional)	AKI staging, fluid plan, hyperkalaemia treatment, dialysis decision. Most hospitals (K^+ at RRH/NRH)
Serum albumin, total protein, cholesterol	Nephrotic	**Albumin <30 g/L**, cholesterol high	Confirms nephrotic syndrome. Hospitals
Urine protein:creatinine ratio (UPCR)	Nephrotic at diagnosis and to confirm remission/relapse	**≥200 mg/mmol** nephrotic; ≤20 remission	Definitions. RRH/NRH (dipstick otherwise)
C3 (± C4)	Haematuria with nephritic features	**Low C3** in PSGN (normal C4); low C3 and C4 in lupus	**C3 still low after 8 weeks** → MPGN/C3 glomerulopathy/lupus → biopsy. NRH/private labs
ASO / anti-DNase B	Suspected PSGN	ASO raised (throat); **anti-DNase B** raised (skin)	Supports PSGN. Hospitals (anti-DNase B referral labs)
Throat swab / skin swab culture	Active infection	GAS	Antibiotics, contact treatment
FBC	All	Anaemia (dilutional, HUS, malaria), high Hct (hypovolaemia), low platelets (HUS, sepsis)	Diagnosis and transfusion
Malaria RDT/film	All febrile children; AKI	Positive	Treat malaria (artesunate; no dose change in AKI)
HIV, hepatitis B surface antigen, VDRL/RPR	Every nephrotic child	Positive	Secondary nephrotic syndrome; ART; avoid steroids in HBV-related disease without specialist input
TB screen (history, CXR, GeneXpert if symptoms; TST where available)	Before long steroid courses	TB	Treat TB first / alongside per specialist
Stool and urine microscopy for parasites	Nephrotic in endemic areas	*Schistosoma*, *Strongyloides*	Praziquantel 40 mg/kg (UCG); albendazole/ivermectin before steroids
Renal ultrasound	AKI, atypical nephrotic, PSGN not resolving	Small kidneys (CKD), hydronephrosis (obstruction), enlarged echogenic kidneys (HIVAN, acute GN), renal vein thrombosis	Changes diagnosis completely. Most hospitals
CXR	Breathless, overload	Pulmonary oedema, cardiomegaly, effusions	Fluid and diuretic decisions
Blood gas / bicarbonate	AKI stage 2–3	Metabolic acidosis	Bicarbonate, dialysis. RRH/NRH
Calcium, phosphate	AKI	Low Ca, high PO_4	Binders, dialysis

Test	When	Expected / abnormal finding	How it changes management
Renal biopsy	**Atypical nephrotic, SRNS, PSGN not resolving** (low C3 >8 weeks, persistent nephrotic proteinuria, rising creatinine), RPGN	FSGS, MPGN, crescents, membranous, lupus	Specific immunosuppression. **Mulago only**

Step 5: Putting it together

Model one-liner (PSGN): "A 9-year-old boy with **cola-coloured urine, periorbital oedema and oliguria** for 3 days, **3 weeks after infected scabies**, with **BP 140/95 (>99th centile)**, urine blood 3+/protein 1+, creatinine 115 µmol/L (about 2.5× expected), **low C3** and raised anti-DNase B: **acute post-streptococcal glomerulonephritis with stage 2 AKI and stage 2 hypertension**, not yet in heart failure."

Model one-liner (nephrotic): "A 4-year-old girl with **2 weeks of generalised oedema**, urine protein **4+**, no haematuria, normal BP, **albumin 14 g/L**, cholesterol 9.8 mmol/L, normal creatinine and C3, HIV and HBsAg negative: **first presentation of idiopathic nephrotic syndrome, typical features**, warm peripheries, no infection."

Worked example: is this AKI and what stage?

A 3-year-old (14 kg, 95 cm) with diarrhoea has creatinine **98 µmol/L**. Expected creatinine at this age is about 30 µmol/L. Schwartz eGFR = 36.5 × 95 ÷ 98 ≈ **35 mL/min/1.73 m²**. Creatinine is **>3 × expected** and eGFR is about 35 → **KDIGO stage 3**. After rehydration, creatinine falls to 45 µmol/L in 24 h: this was mainly **pre-renal** AKI.

Worked example: daily fluid allowance in oliguric PSGN

Child 25 kg, 125 cm: BSA (Mosteller) = √(125 × 25 ÷ 3600) ≈ **0.93 m²**. Insensible losses ≈ **400 mL/m²/day** ≈ 370 mL/day. Yesterday's urine output = 300 mL. **Today's total intake (oral + IV) ≈ 370 + 300 = 670 mL** (≈28 mL/h), adjusted by daily weight. (UCG 2023 simplifies this as **10 mL/kg/day + measured losses**, which is similar in a school-age child but too little in infants.)

Differential diagnosis

Condition	Distinguishing features	Key investigation
PSGN	After strep throat/skin, low C3 recovering by 8 weeks, good prognosis	C3, ASO/anti-DNase B
IgA nephropathy	Gross haematuria **during** (1–2 days after) an URTI, normal C3	Biopsy
HSP (IgA vasculitis) nephritis	Purpura on legs/buttocks, arthritis, abdominal pain	Clinical; urine
Lupus nephritis	Adolescent girl, rash, arthritis, cytopenias, low C3 and C4	ANA, dsDNA, biopsy
MPGN / C3 glomerulopathy	Nephritic-nephrotic, persistent low C3	Biopsy
Haemolytic uraemic syndrome	Bloody diarrhoea, pallor, low platelets, fragmented RBCs, AKI	FBC, film
Minimal change disease	Age 1–6, pure nephrotic, steroid-sensitive	Response to steroids
FSGS	Older, haematuria/hypertension, steroid-resistant; HIVAN in HIV	Biopsy, HIV
Membranous (HBV)	Nephrotic in HBsAg-positive child	HBsAg, biopsy
Quartan malaria / schistosomal nephropathy	Endemic exposure, steroid-resistant	Film, stool/urine microscopy, biopsy

Condition	Distinguishing features	Key investigation
Kwashiorkor	Oedema with malnutrition, skin/hair changes, **no heavy proteinuria**	Urine dipstick, nutrition history
Heart failure, liver disease	Hepatomegaly, gallop, jaundice, no proteinuria	Echo, LFTs
Haemoglobinuria (malaria, G6PD)	Dark urine, anaemia, jaundice, no RBCs in urine	Microscopy, Hb, RDT

Investigations

Test	What you are looking for	Why
Dipstick + microscopy	Blood, protein, casts	Nephritic vs nephrotic vs haemoglobinuria
Creatinine, urea, electrolytes	AKI, K^+, Na^+	Staging, emergencies
Albumin, cholesterol, UPCR	Nephrotic definitions	Diagnosis, remission, relapse
C3, ASO/anti-DNase B	PSGN	Diagnosis; follow-up C3 at 8 weeks
FBC, malaria test	Anaemia, HUS, malaria	Causes and complications
HIV, HBsAg, syphilis, TB screen	Secondary causes, steroid safety	Changes treatment
Renal ultrasound	Size, obstruction, echogenicity	AKI cause, CKD
Biopsy (Mulago)	Histology	Atypical or resistant disease

Management

A. Acute post-streptococcal glomerulonephritis

Algorithm

1. **Admit** (UCG: refer oedematous, hypertensive, oliguric children to hospital/RRH). **Daily weight, strict input/output, BP 4–6-hourly** (hourly if severe).
2. **Fluid restriction**: if oliguric or overloaded, **insensible losses (~400 mL/m²/day) + previous day's urine output** (UCG: 10 mL/kg + losses). No maintenance IV fluids. Oral where possible.
3. **Salt restriction**: no added salt (UCG <2 g/day; about half a teaspoon). **Restrict potassium** (bananas, oranges, beans, meat broths) if oliguric; moderate protein, adequate calories.
4. **Furosemide** for oedema, hypertension or congestion.
5. **Control BP** (table below).
6. **Treat the streptococcus**: 10 days of penicillin/amoxicillin (or BPG once). This does **not** alter the nephritis, but clears the nephritogenic strain and protects contacts. Treat **scabies** (and family) and impetigo.
7. **Monitor creatinine and K^+** daily (more often if rising) and treat hyperkalaemia; refer for dialysis if needed.
8. **No steroids** in typical PSGN.
9. **Follow-up**: BP and urine at 2 weeks, 6 weeks, **C3 at 8 weeks**, then 3, 6 and 12 months (Medscape) until urine and BP are normal for a year.

Drug	Dose	Route	Frequency	Max	Notes
Furosemide	**1–2 mg/kg** (UCG: 1 mg/kg every 8–12 h; up to 2 mg/kg IV for marked congestion)	Oral or IV	8–12-hourly	**6 mg/kg/day**	Avoid large doses in anuria (ototoxic, ineffective)

Drug	Dose	Route	Frequency	Max	Notes
Nifedipine (short-acting)	**0.25–0.5 mg/kg/dose**	Oral (swallowed, not sublingual)	6–8-hourly as needed	**10 mg/dose**, 3 mg/kg/day	Widely available in Uganda; UCG: child BP treatment by specialist. Avoid big, rapid falls
Amlodipine	**0.1–0.2 mg/kg/day** (start 0.1)	Oral	Once daily	**10 mg/day**	Good for persistent hypertension
Hydralazine	**0.1–0.2 mg/kg** (max 20 mg)	IV/IM	Every 4–6 h	3.5 mg/kg/day	For severe hypertension when oral not possible
Labetalol (where available)	Infusion **0.25–3 mg/kg/h** (or 0.2–1 mg/kg bolus)	IV	Continuous	3 mg/kg/h	Hypertensive emergency; avoid in pulmonary oedema with poor LV, asthma
Avoid	ACE inhibitors/ARBs in the acute oliguric phase	—	—	—	Hyperkalaemia, worsen GFR
Penicillin V	**10–20 mg/kg/dose** (UCG; 250 mg <30 kg, 500 mg ≥30 kg)	Oral	6-hourly	—	**10 days**
or **Amoxicillin**	**20 mg/kg/dose** (UCG) or 50 mg/kg once daily (max 1 g)	Oral	8-hourly (or daily)	—	10 days
or **Benzathine penicillin**	**0.6 MU <30 kg; 1.2 MU ≥30 kg**	Deep IM	Once	—	Single dose; good adherence
Penicillin allergy: **erythromycin**	**15 mg/kg/dose** (UCG) (max 500 mg)	Oral	6-hourly	—	10 days

RED FLAG

Hypertensive emergency (encephalopathy, seizures, visual loss, heart failure): lower BP **by no more than 25% of the planned reduction in the first 8 hours**, then to normal over 24–48 hours, using IV labetalol, IV hydralazine or oral nifedipine, plus **IV furosemide**. Too fast a fall can cause cerebral ischaemia and blindness. Treat seizures with benzodiazepines and correct sodium and glucose.

Prognosis: >95% of children recover fully. **Gross haematuria** clears in 1–3 weeks, **oedema and hypertension** within 1–3 weeks, **C3** normalises by **6–8 weeks**, **microscopic haematuria** may persist for **up to a year** and proteinuria for up to 6 months. Recurrence is rare (type-specific immunity).

When to refer for biopsy / nephrology: C3 low beyond 8 weeks, **nephrotic-range proteinuria** persisting, **creatinine rising** or oliguria beyond 1 week, **hypertension beyond 6 weeks**, gross haematuria beyond 3 weeks, no evidence of streptococcal infection, systemic features (rash, arthritis), or a family history of kidney disease.

B. Nephrotic syndrome

Initial assessment and general care (first presentation: admit)

1. **Classify** as typical (age 1–12, normal BP and creatinine, no gross haematuria, normal C3) or **atypical** (refer for biopsy before steroids).
2. **Screen**: HIV, HBsAg, syphilis, **TB** (history/CXR), malaria, stool/urine parasites. **Deworm** (albendazole 400 mg; ivermectin if *Strongyloides* risk) and **praziquantel 40 mg/kg** if schistosomiasis endemic (UCG).

3. **Salt restriction** (<2 g/day, UCG). **Fluid**: restrict only if hyponatraemic or very oedematous (RCH: 50–70% of maintenance); otherwise normal intake. Normal protein for age (no high-protein diet).
4. **Oedema**: mild to moderate oedema needs only salt restriction and steroids. Severe/symptomatic oedema (respiratory compromise, scrotal skin breakdown) → **albumin 20% 0.5–1 g/kg (2.5–5 mL/kg) over 4 h with furosemide 0.5–1 mg/kg IV (max 40 mg)** mid-way or at the end (RCH). Use **furosemide alone only if the child is clearly not hypovolaemic** (warm, normal CRT, no tachycardia, normal Hct).
5. **Hypovolaemia** (cold peripheries, tachycardia, abdominal pain, CRT >3 s, rising Hct, urine Na <10 mmol/L): **albumin 20% 1 g/kg** (or 4.5% albumin/crystalloid 10–20 mL/kg) and **stop diuretics**.
6. **Prednisolone** (below).
7. **Infection**: treat promptly (peritonitis: **ceftriaxone 80–100 mg/kg/day**). Routine penicillin prophylaxis is **not** recommended by IPNA/KDIGO, though some African units give it while oedematous (local policy).
8. **Thrombosis prevention**: mobilise, avoid over-diuresis and femoral lines; anticoagulation only for previous thrombosis or very high risk (specialist).
9. **Vaccines**: complete **PCV**; give **PPSV23**; influenza yearly; **avoid live vaccines** (measles, varicella, BCG, OPV, yellow fever) while on high-dose steroids (≥2 mg/kg/day or ≥20 mg/day for ≥14 days) until **1 month after stopping**.
10. **Education**: home urine dipstick (first morning urine daily during relapses and when unwell; weekly otherwise), steroid card, signs of relapse, infection and thrombosis.

Prednisolone regimens

Situation	Dose	Frequency	Duration	Source
First episode (current international)	**60 mg/m²/day or 2 mg/kg/day (max 60 mg)**	**Once daily** in the morning	**4 weeks**	IPNA 2023; KDIGO 2021/2025; RCH
then	**40 mg/m² or 1.5 mg/kg (max 40 mg)**	**Alternate days**	**4 weeks**, then stop (total 8 weeks; KDIGO allows 8–12 weeks, e.g. 6 + 6)	IPNA 2023; KDIGO
First episode (UCG 2023 regimen)	2 mg/kg daily (max 60 mg)	Daily	Until proteinuria clears (about 6 weeks); taper from week 4 by 0.5 mg/kg/day each week	UCG 2023
Relapse	**60 mg/m² or 2 mg/kg (max 60 mg)**	Daily	**Until remission for 3 consecutive days** (protein negative/trace)	IPNA/KDIGO
then	**40 mg/m² or 1.5 mg/kg (max 40 mg)**	Alternate days	**4 weeks**, then stop or taper	IPNA/KDIGO
FRNS / SDNS	Steroid-sparing agents (specialist): **levamisole 2.5 mg/kg alternate days**, **oral cyclophosphamide 2 mg/kg/day for 12 weeks** (cumulative ≤168 mg/kg), mycophenolate, calcineurin inhibitors, rituximab	—	Months to years	KDIGO/IPNA; Mulago
SRNS	Refer: **biopsy**, genetic testing if available, **ACE inhibitor** (e.g. enalapril) for proteinuria, **calcineurin inhibitor** (ciclosporin/tacrolimus) first line	—	—	IPNA 2020 SRNS; KDIGO

UGANDA CONTEXT

Many Ugandan units, including some at Mulago, still use a **6 weeks daily + 6 weeks alternate-day** regimen or the UCG taper. **IPNA 2023 and KDIGO 2025** recommend **4 weeks daily + 4 weeks alternate-day** for typical first episodes (longer courses do not reduce later frequent relapses and add toxicity), while KDIGO accepts 8–12 weeks in total. **Follow your unit's protocol and know both for the exam.** In practice, round doses to tablet sizes (5 mg tablets) and give them as **one morning dose** with food.

Steroid side effects to watch: hypertension, hyperglycaemia, Cushingoid face, weight gain, mood/behaviour change, gastritis (give with food), infection, growth suppression, cataract/glaucoma (long term), adrenal suppression after prolonged courses (stress dosing when ill).

C. Acute kidney injury: step by step

1. **Recognise and stage** (creatinine vs expected for age; urine output).
2. **Find and treat the cause** while protecting the kidney:
   - **Hypovolaemic** (dry, shocked, diarrhoea): **10–20 mL/kg Ringer's lactate or NS** over 15–60 min (careful in SAM and severe anaemia, per WHO), reassess, repeat once if needed, then rehydrate. **Urine output should improve** within hours if pre-renal.
   - **Euvolaemic or overloaded**: **restrict** to insensible losses + urine output + other losses. **Furosemide 1–2 mg/kg IV** may convert oliguric to non-oliguric AKI (it does not improve recovery); if no response after 2 doses, stop.
   - **Obstruction**: catheterise (infant boy with PUV: feeding tube catheter), urgent ultrasound and surgical referral.
   - **Infection/malaria**: ceftriaxone (no adjustment needed) and **IV artesunate** (no adjustment); treat sepsis.
3. **Stop nephrotoxins**: NSAIDs, aminoglycosides (if essential, give **gentamicin with extended dosing interval and levels**), ACE inhibitors, contrast, herbal medicines.
4. **Adjust drug doses** to eGFR (penicillins, cotrimoxazole, aciclovir, digoxin, morphine).
5. **Monitor**: weight twice daily, strict input/output, BP, **K^+, Na^+, creatinine, bicarbonate, Ca, PO_4** daily or more often.
6. **Electrolytes and acid–base** (table).
7. **Nutrition**: adequate calories (carbohydrate and fat), **protein ~1–1.5 g/kg/day** (do not starve), **low K^+, low phosphate, low salt**.
8. **Refer for dialysis early** (AEIOU) before the child is moribund (UCG 2023: "as soon as possible and before the patient's condition becomes critical").

Problem	Treatment	Dose
Hyperkalaemia (K^+ ≥6.0 mmol/L; urgent if ≥6.5 or ECG changes)	Stop all K^+ (IV fluids, foods, drugs)	—
	Calcium gluconate 10% (stabilises the heart; does not lower K^+)	**0.5 mL/kg IV over 5–10 min (max 20 mL)** with ECG monitoring; repeat if ECG changes persist
	Salbutamol (shifts K^+ into cells)	**Nebulised 2.5 mg (<25 kg) or 5 mg (≥25 kg)**; repeat every 2 h as needed
	Insulin + glucose	**Soluble insulin 0.1 unit/kg + glucose 0.5 g/kg (5 mL/kg of 10% dextrose)** IV over 30 min; check glucose hourly
	Sodium bicarbonate (if acidotic)	**1–2 mmol/kg IV** over 30 min (8.4% diluted)

Problem	Treatment	Dose
	Calcium resonium / sodium polystyrene (removes K^+)	**1 g/kg/day in divided doses** oral or rectal (max 60 g/day)
	Dialysis	Refractory hyperkalaemia
Hyponatraemia (usually dilutional)	Fluid restriction	Insensible + output
	Seizure/encephalopathy with Na <125	**3% saline 2 mL/kg (max 100 mL) IV over 10 min**, repeat once if needed; raise Na by no more than 8–10 mmol/L in 24 h
Metabolic acidosis	Treat cause; bicarbonate if pH <7.15 or HCO_3^- <12	1–2 mmol/kg IV; oral bicarbonate in chronic
Hyperphosphataemia / hypocalcaemia	Phosphate binder with meals (calcium carbonate); IV calcium only if symptomatic	Calcium gluconate 10% 0.5 mL/kg slowly
Hypertension	Fluid/salt restriction, furosemide, amlodipine/nifedipine/hydralazine	As in PSGN table
Anaemia	Transfuse packed cells slowly if Hb <7 g/dL or symptomatic	5–10 mL/kg over 3–4 h with furosemide (beware K^+ in old blood; ask for fresh units)

D. Dialysis indications (refer early)

Indication	Examples
Acidosis	Refractory metabolic acidosis (pH <7.15 despite treatment)
Electrolytes	**K^+ ≥7 mmol/L** or rising despite medical treatment; severe hyponatraemia/hypernatraemia not correctable
Intoxication	Dialysable poisons (e.g. salicylate, lithium, methanol)
Overload	**Pulmonary oedema** or hypertension not responding to diuretics; **fluid overload >10–20% of body weight**
Uraemia	Encephalopathy, pericarditis, uraemic bleeding; urea very high (e.g. >40 mmol/L, rising)
Other	**Anuria >24 h** (or oliguria with rising creatinine), need for "space" for nutrition, drugs or blood, TLS

Peritoneal dialysis is the practical acute modality for children in Uganda (no vascular access needed, gentle fluid removal); it is available at Mulago paediatric nephrology. Haemodialysis is limited for small children.

> **DRUG DOSING**
> **Gentamicin in AKI**: avoid if possible. If essential (e.g. neonatal sepsis), give the normal **first dose** (7.5 mg/kg in children per WHO) and **extend the interval** (e.g. every 36–48 h) guided by trough levels (<1 mg/L) where available. Never repeat a nephrotoxic dose "by the clock" in an anuric child.

Complications

PSGN	Nephrotic syndrome	AKI
Hypertensive encephalopathy, PRES, seizures	Infection: **SBP** (pneumococcus), cellulitis, sepsis	Hyperkalaemia, arrhythmia
Pulmonary oedema, heart failure	**Thrombosis** (renal vein, cerebral venous sinus, PE)	Pulmonary oedema
AKI, hyperkalaemia	Hypovolaemia, shock, AKI	Metabolic acidosis

PSGN	Nephrotic syndrome	AKI
Rapidly progressive GN (crescents, rare)	Hyperlipidaemia	Uraemic encephalopathy, bleeding
Rare progression to CKD	Malnutrition, growth failure	Drug toxicity
	Steroid toxicity; relapses; SRNS → CKD	Progression to CKD (follow-up BP and urine after any AKI)

Ward-round & exam pearls

Model case presentation

"**Mukisa is a 4-year-old boy weighing 18.5 kg** today, up from **16 kg** a month ago (oedema; length 101 cm; pre-illness weight-for-height Z 0, MUAC unreliable with oedema). He is **fully immunised per UNEPI** including PCV and measles-rubella, eats family foods, and is **HIV-negative** (mother negative; HBsAg negative). He presented with **10 days of progressive swelling**, starting around the eyes in the morning and now in the abdomen, legs and scrotum, with frothy urine and reduced urine output. There is **no** haematuria, sore throat or skin infection. On examination he has generalised pitting oedema, ascites and scrotal oedema; he is **warm with CRT <2 s**, HR 108, **BP 96/58 (normal for height)**, and the abdomen is non-tender. Urine protein **4+**, blood negative. **Albumin 13 g/L**, cholesterol 10.2 mmol/L, creatinine 28 µmol/L, Na 131, K 4.2. **C3 normal.** This is a **first presentation of typical idiopathic nephrotic syndrome**, probably minimal change disease, without complications. I have screened for TB (no contact, normal CXR) and given albendazole. Plan: **prednisolone 60 mg/m²/day** calculated on his **pre-oedema weight of 16 kg** (BSA ≈ 0.67 m² → **40 mg once daily**; 2 mg/kg would give 32 mg; oedema weight overestimates the dose) for 4 weeks, then 40 mg/m² (≈25–27.5 mg, rounded to tablet size) on alternate days for 4 weeks, no-added-salt diet, daily dipstick, weight and BP, and education for home monitoring."

Viva questions (with answers)

1. **Why is C3 low in PSGN and when should it normalise?** Immune complexes activate the alternative complement pathway; C3 normalises within 6–8 weeks; if not, think MPGN/C3 glomerulopathy/lupus and biopsy.
2. **Why do antibiotics not cure PSGN?** The nephritis is immune-complex mediated after the infection; antibiotics only eradicate the strain and protect contacts.
3. **How do you tell an underfilled from an overfilled nephrotic child?** Underfilled: tachycardia, cold peripheries, CRT >3 s, abdominal pain, postural hypotension, high Hct, urine Na <10–20 mmol/L. Overfilled: hypertension, crackles, raised JVP, no tachycardia.
4. **Define steroid-dependent nephrotic syndrome.** Two consecutive relapses during steroid therapy or within 14 days of stopping.
5. **Give the KDIGO urine output criteria for AKI stage 3.** <0.3 mL/kg/h for ≥24 h or anuria for ≥12 h.
6. **What are the indications for dialysis?** AEIOU: refractory acidosis, electrolytes (K^+), intoxication, overload, uraemia; plus prolonged anuria.

OSCE tips

- **Urine dipstick station**: check expiry, fresh specimen, dip briefly, read each pad at the stated time, and interpret: blood + protein (nephritic), protein 3–4+ (nephrotic), blood without RBCs (haemoglobinuria).
- **BP station**: choose the cuff by **bladder width ~40% of mid-arm circumference** and length 80–100%; a small cuff overestimates BP.
- **Counselling for nephrotic syndrome**: relapses are common (70–80%), home dipstick, steroid side effects, when to come back (fever, abdominal pain, swelling, reduced urine), avoid live vaccines on steroids, no added

salt.

Mnemonics & memory aids

- **Nephritic "PHAROH"**: **P**roteinuria (mild), **H**aematuria, **A**zotaemia, **R**BC casts, **O**liguria, **H**ypertension.
- **Nephrotic "PALE"**: **P**roteinuria (massive), **A**lbumin low, **L**ipids high, **E**dema.
- **PSGN timing "1–2 throat, 3–6 skin"** (weeks).
- **Steroid regimen "60 then 40, 4 and 4"** (mg/m²; weeks) for IPNA/KDIGO first episode.
- **Dialysis "AEIOU"**.
- **Hyperkalaemia "C BIG K Drop"**: **C**alcium, **B**icarbonate/β-agonist (salbutamol), **I**nsulin + **G**lucose, **K**ayexalate/resonium, **D**ialysis.

High-yield summary

Numbers & doses to know

Item	Value
PSGN latency	1–2 weeks (throat); 3–6 weeks (skin)
C3 normalises	By 6–8 weeks
Nephrotic proteinuria	UPCR ≥200 mg/mmol; dipstick ≥3+; ≥40 mg/m²/h
Nephrotic albumin	<30 g/L (IPNA/KDIGO; <25 g/L older/RCH)
Remission	Negative/trace (UPCR ≤20) × 3 days
Relapse	≥3+ (UPCR ≥200) × 3 days
SRNS	No remission after 4 weeks full dose
FRNS	≥2 in 6 months after initial response or ≥4 in 12 months
Prednisolone first episode	60 mg/m² (2 mg/kg, max 60) daily × 4 wk → 40 mg/m² (1.5 mg/kg, max 40) alt days × 4 wk
Albumin 20%	0.5–1 g/kg (2.5–5 mL/kg) over 4 h + furosemide 0.5–1 mg/kg (max 40 mg)
Insensible losses	~400 mL/m²/day (UCG: 10 mL/kg/day)
Furosemide	1–2 mg/kg; max 6 mg/kg/day
Nifedipine	0.25–0.5 mg/kg/dose (max 10 mg)
Amlodipine	0.1–0.2 mg/kg/day (max 10 mg)
Hydralazine IV	0.1–0.2 mg/kg (max 20 mg)
BP lowering in emergency	≤25% of planned reduction in first 8 h
KDIGO stage 1	Cr 1.5–1.9× or +26.5 µmol/L in 48 h; UO <0.5 mL/kg/h 6–12 h
KDIGO stage 3	Cr ≥3× or ≥353.6 µmol/L or dialysis or eGFR <35 (<18 y); UO <0.3 × 24 h or anuria 12 h
Schwartz eGFR	36.5 × height (cm) ÷ creatinine (µmol/L)
Calcium gluconate 10%	0.5 mL/kg IV (max 20 mL)
Insulin/glucose	0.1 U/kg + 0.5 g/kg glucose
3% saline for hyponatraemic seizures	2 mL/kg (max 100 mL)

Classic exam traps

- **Trap**: Maintenance IV fluids for an oliguric PSGN child "to flush the kidneys". **Right**: restrict to insensible losses + output.
- **Trap**: Steroids for PSGN. **Right**: not indicated in typical PSGN.
- **Trap**: Furosemide for a cold, tachycardic nephrotic child with abdominal pain. **Right**: this is hypovolaemia; give albumin and stop diuretics.
- **Trap**: Giving IV calcium gluconate expecting it to lower K^+. **Right**: it only protects the heart; you must also shift and remove K^+.
- **Trap**: Using an adult creatinine range for a toddler. **Right**: 70 µmol/L can be stage 2–3 AKI in a 2-year-old.
- **Trap**: Measles vaccine for a child on 2 mg/kg/day prednisolone. **Right**: wait until 1 month off high-dose steroids.
- **Trap**: Starting prednisolone without checking TB, HIV, hepatitis B and worms. **Right**: screen and deworm first.
- **Trap**: Reading dipstick "blood 3+" as haematuria in a child with malaria. **Right**: check microscopy for RBCs; it may be haemoglobinuria.

Questions seniors ask

1. **Nephritic or nephrotic, and why?** Name the features (haematuria, hypertension, oliguria vs heavy proteinuria, hypoalbuminaemia, oedema).
2. **What is this child's fluid allowance today?** Insensible (~400 mL/m²/day) + yesterday's urine output + other losses, guided by weight.
3. **When will you biopsy a child with nephrotic syndrome?** Atypical features (age <1 or >12, persistent hypertension, gross haematuria, low C3, raised creatinine, secondary cause), steroid resistance, before calcineurin inhibitors.
4. **How do you prevent infection in a nephrotic child?** Vaccination (PCV/PPSV23, influenza; live vaccines when off high-dose steroids), early treatment of fever/abdominal pain, education; routine antibiotic prophylaxis is not recommended by IPNA/KDIGO.
5. **What is the commonest cause of AKI on our ward and how do you prevent it?** Dehydration from diarrhoea and severe malaria; early rehydration, avoiding nephrotoxins, monitoring urine output.
6. **What do you do first with K^+ 7.2 and peaked T waves?** Calcium gluconate 10% 0.5 mL/kg IV over 5–10 min with ECG monitoring, then salbutamol and insulin–glucose, stop K^+, arrange dialysis.
7. **What follow-up does a child need after PSGN?** BP and urine at 2 and 6 weeks, C3 at 8 weeks, then periodically until haematuria, proteinuria and hypertension have gone for a year.

SELF-TEST

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Features of nephritic syndrome?	**Haematuria** (cola urine, RBC casts), **hypertension**, **oedema** (periorbital), **oliguria**, mild proteinuria, raised creatinine. (PHAROH)
2	Features of nephrotic syndrome?	**Nephrotic-range proteinuria** (UPCR ≥200 mg/mmol, dipstick ≥3+), **hypoalbuminaemia** (<30 g/L IPNA/KDIGO), **oedema**, **hyperlipidaemia**. (PALE)
3	Latency of PSGN after throat vs skin infection?	**1–2 weeks** after pharyngitis; **3–6 weeks** after skin infection (impetigo, infected scabies).
4	Complement in PSGN?	**Low C3** (alternative pathway), usually normal C4; C3 **normalises within 6–8 weeks**. Persistently low beyond 8 weeks → MPGN/C3 glomerulopathy/lupus → biopsy.
5	Which antibody test supports PSGN after skin infection?	**Anti-DNase B** (ASO is often not raised after skin infection because skin lipids inactivate streptolysin O).
6	Fluid allowance in oliguric PSGN/AKI?	**Insensible losses (~400 mL/m²/day) + previous day's urine output + other losses**, adjusted by daily weight. (UCG simplifies: 10 mL/kg/day + losses.)
7	Do antibiotics change the course of PSGN?	No. They eradicate the nephritogenic strain (penicillin/amoxicillin 10 days or BPG once) and protect contacts; the nephritis is immune-mediated.
8	Hypertensive emergency in children: how fast do you lower BP?	By **no more than 25% of the planned reduction in the first 8 hours**, then to target over 24–48 h (IV labetalol/hydralazine or oral nifedipine, plus furosemide).
9	Child antihypertensive doses: nifedipine, amlodipine, hydralazine?	**Nifedipine 0.25–0.5 mg/kg/dose** (max 10 mg); **amlodipine 0.1–0.2 mg/kg/day** (max 10 mg); **hydralazine IV 0.1–0.2 mg/kg** (max 20 mg) every 4–6 h.
10	When should a child with PSGN be referred for biopsy/nephrology?	C3 low **>8 weeks**, persistent **nephrotic-range proteinuria**, **rising creatinine**/oliguria >1 week, hypertension >6 weeks, gross haematuria >3 weeks, no evidence of strep, systemic features.
11	Definition of remission and relapse in nephrotic syndrome?	**Remission**: dipstick negative/trace (UPCR ≤20 mg/mmol) for **3 consecutive days**. **Relapse**: dipstick **≥3+** (UPCR ≥200 mg/mmol) for **3 consecutive days**.
12	Define steroid-resistant nephrotic syndrome.	**No complete remission after 4 weeks** of standard-dose prednisolone (60 mg/m²/day). IPNA adds a confirmation period of up to 2 weeks. Refer for biopsy, genetics, RAS blockade and calcineurin inhibitor.
13	Define frequently relapsing and steroid-dependent nephrotic syndrome.	**FRNS**: ≥2 relapses within 6 months of initial response, or ≥4 in any 12 months. **SDNS**: 2 consecutive relapses during steroid therapy or within **14 days** of stopping.
14	First-episode prednisolone regimen (IPNA 2023 / KDIGO)?	**60 mg/m²/day or 2 mg/kg/day (max 60 mg)** once daily for **4 weeks**, then **40 mg/m² or 1.5 mg/kg (max 40 mg) alternate days** for **4 weeks** (KDIGO: 8–12 weeks total).
15	Relapse prednisolone regimen?	**60 mg/m² (2 mg/kg, max 60 mg) daily until remission for 3 days**, then **40 mg/m² (1.5 mg/kg, max 40 mg) alternate days for 4 weeks**.

#	Question	Answer
16	Signs that a nephrotic child is hypovolaemic (underfilled)?	**Tachycardia, cold peripheries, CRT >3 s, abdominal pain, postural hypotension, oliguria, rising haematocrit, urine Na <10–20 mmol/L**. Stop diuretics; give albumin.
17	Albumin infusion dose for severe nephrotic oedema?	**20% albumin 0.5–1 g/kg (2.5–5 mL/kg) over 4 h** with **furosemide 0.5–1 mg/kg IV (max 40 mg)** mid-way or at the end (RCH).
18	Screening before starting prednisolone in a Ugandan child with nephrotic syndrome?	**TB** (history, CXR ± Xpert), **HIV, hepatitis B, syphilis**, malaria, stool/urine for **schistosomiasis/Strongyloides**; deworm (albendazole) and praziquantel 40 mg/kg if endemic.
19	Secondary causes of nephrotic syndrome relevant to Uganda?	**HIV-associated nephropathy**, **hepatitis B** (membranous), **quartan malaria** (*P. malariae*), **schistosomiasis** (MPGN), **syphilis**, PSGN, SLE, drugs.
20	KDIGO AKI stage 1 criteria?	Creatinine **1.5–1.9 × baseline** (within 7 days) or rise **≥26.5 µmol/L within 48 h**; or urine output **<0.5 mL/kg/h for 6–12 h**.
21	KDIGO AKI stage 3 criteria?	Creatinine **≥3 × baseline** or **≥353.6 µmol/L** or **dialysis** or **eGFR <35 mL/min/1.73 m² in under-18s**; or urine output **<0.3 mL/kg/h for ≥24 h** or **anuria ≥12 h**.
22	Bedside Schwartz formula?	**eGFR = 36.5 × height (cm) ÷ creatinine (µmol/L)** (or 0.413 × height ÷ creatinine in mg/dL).
23	Emergency treatment of hyperkalaemia in a child?	**Calcium gluconate 10% 0.5 mL/kg (max 20 mL) IV over 5–10 min**; **salbutamol neb 2.5 mg (<25 kg) / 5 mg**; **insulin 0.1 U/kg + glucose 0.5 g/kg**; bicarbonate 1–2 mmol/kg if acidotic; resonium 1 g/kg/day; dialysis.
24	Dialysis indications?	**AEIOU**: refractory **A**cidosis, **E**lectrolytes (K^+ ≥7 or refractory), **I**ntoxication, **O**verload (pulmonary oedema, >10–20% fluid overload), **U**raemia (encephalopathy, pericarditis, bleeding); prolonged anuria.
25	Pre-renal vs intrinsic AKI on urine tests?	**Pre-renal**: urine Na <20 mmol/L, **FENa <1%**, concentrated urine, bland sediment. **Intrinsic (ATN)**: urine Na >40, **FENa >2%**, granular casts. (FENa unreliable after furosemide.)
26	Live vaccines and steroids in nephrotic syndrome?	Avoid live vaccines (measles, varicella, BCG, OPV, yellow fever) on **≥2 mg/kg/day or ≥20 mg/day for ≥14 days** until **1 month after stopping**. Give PCV/PPSV23 and annual influenza.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 9-year-old boy (26 kg) has cola-coloured urine, puffy eyes and reduced urine for 3 days. He had infected scabies 4 weeks ago. BP 142/96 mmHg (>99th centile), no crackles, liver not enlarged. Urine: blood 3+, protein 1+. Creatinine 104 µmol/L, K^+ 5.1. What is the MOST appropriate initial management?

A IV normal saline at maintenance rate to improve kidney perfusion

B Prednisolone 2 mg/kg/day

C Fluid restriction to insensible losses plus urine output, no added salt, furosemide 1–2 mg/kg, antihypertensive (e.g. nifedipine or amlodipine), and penicillin for 10 days

D Enalapril 0.5 mg/kg/day as first-line antihypertensive

E Discharge with oral amoxicillin and review in 2 weeks

2. A 3-year-old girl (14 kg) has had generalised swelling for 10 days. BP 94/56, CRT <2 s, warm peripheries. Urine protein 4+, blood negative. Albumin 14 g/L, cholesterol 9 mmol/L, creatinine 25 µmol/L, C3 normal. HIV and HBsAg negative; no TB contact; CXR normal. What is the MOST appropriate treatment?

A Furosemide 2 mg/kg twice daily alone

B Renal biopsy before any treatment

C Cyclophosphamide 2 mg/kg/day

D Albumin infusion daily until albumin is above 30 g/L

E Prednisolone 60 mg/m²/day (2 mg/kg/day, max 60 mg) once daily for 4 weeks, then 40 mg/m² on alternate days for 4 weeks, with a no-added-salt diet

3. A 5-year-old (17 kg) with nephrotic syndrome on day 5 of prednisolone was given furosemide 2 mg/kg twice daily by a junior doctor. He now has abdominal pain, HR 152, cold hands, CRT 4 s, BP 88/50, and urine output 0.3 mL/kg/h. Hct has risen from 38% to 47%. Abdomen soft, no fever. What is the BEST next step? Advanced

A Stop furosemide and give 20% albumin 1 g/kg (or 10–20 mL/kg crystalloid/4.5% albumin if unavailable), then reassess

B Increase furosemide to clear the oedema

C Start ceftriaxone and arrange laparotomy

D Give a 40 mL/kg saline bolus over 10 minutes

E Restrict fluids to insensible losses

4. A 2-year-old (11 kg, 85 cm) with 3 days of watery diarrhoea is lethargic with sunken eyes, CRT 3 s and no urine for 12 hours. Creatinine is 96 µmol/L (expected about 25–30). Which statement is CORRECT?

A Creatinine is in the adult normal range, so there is no AKI

B He has AKI stage 1 only

C He needs dialysis immediately

D He has KDIGO stage 3 AKI (creatinine >3× expected; anuria ≥12 h), most likely pre-renal, and needs rehydration followed by reassessment of urine output and creatinine

E Give furosemide 2 mg/kg first to restart urine flow

5. An 8-year-old (24 kg) with PSGN has K^+ 7.0 mmol/L and the ECG shows peaked T waves and a widening QRS. What is the FIRST drug to give?

A Calcium resonium 1 g/kg orally
B Calcium gluconate 10% 12 mL (0.5 mL/kg) IV over 5–10 minutes with ECG monitoring
C Furosemide 1 mg/kg IV
D Sodium bicarbonate 1 mmol/kg IV
E Salbutamol 2.5 mg nebulised and repeat the K^+ in 6 hours

6. An 11-year-old girl with PSGN is reviewed 10 weeks after presentation. BP is normal and oedema has resolved, but she still has microscopic haematuria and her C3 remains low. What is the MOST appropriate next step?

Advanced

A Reassure; persistent low C3 is expected for 6 months
B Start prednisolone
C Refer to paediatric nephrology for evaluation including renal biopsy (possible MPGN, C3 glomerulopathy or lupus)
D Repeat a 10-day penicillin course
E Start ciclosporin

7. A 6-year-old (20 kg) with steroid-sensitive nephrotic syndrome in remission is due her measles-rubella second dose. She completed prednisolone 2 mg/kg/day for 4 weeks then alternate-day steroid 2 weeks ago. What is the BEST advice?

A Delay the live vaccine until at least 1 month after stopping high-dose steroids
B Give it today since she is in remission
C Never give live vaccines to children who have had nephrotic syndrome
D Give it only if her albumin is above 40 g/L
E Give immunoglobulin instead of the vaccine

8. A 4-year-old with steroid-sensitive nephrotic syndrome has dipstick 3+ proteinuria on 3 consecutive mornings, 5 months after completing her first steroid course. This is her first relapse. Weight 16 kg. What is the MOST appropriate treatment?

A Repeat the full 8-week first-episode regimen
B Start cyclophosphamide
C Observe without treatment since she is not oedematous
D Renal biopsy
E Prednisolone 2 mg/kg/day (32 mg; or 60 mg/m²) daily until protein-free for 3 days, then 1.5 mg/kg (24 mg; or 40 mg/m²) on alternate days for 4 weeks

9. A 7-year-old with nephrotic syndrome on prednisolone has fever 38.9 °C, diffuse abdominal pain and tenderness with guarding. HR 140, CRT 2 s. What is the MOST likely diagnosis and action?

A Steroid-induced gastritis; give omeprazole
B Hypovolaemia; give albumin only
C Appendicitis; arrange laparotomy before antibiotics
D Spontaneous bacterial peritonitis (usually pneumococcal); blood culture (± ascitic tap) and IV ceftriaxone
E Renal vein thrombosis; start heparin

10. A 12-year-old girl (35 kg) has had nephrotic syndrome for 3 weeks, with haematuria, BP 138/90 and creatinine 98 µmol/L. She is HIV-positive on ART. What is the BEST next step? Advanced

A Start standard prednisolone 60 mg/m² and review in 4 weeks

B Refer to paediatric nephrology for renal biopsy and specialist care (atypical nephrotic syndrome, possible HIVAN/FSGS)

C Stop her ART

D Give albumin and furosemide only and discharge

E Start cyclophosphamide

11. A 10-year-old (30 kg) with PSGN suddenly develops headache, vomiting, blurred vision and a generalised seizure. BP 170/115 (well above the 99th centile + 5 mmHg). Glucose 5.2, Na 134. The seizure stops after diazepam. What is the BEST BP plan? Advanced

A Give sublingual nifedipine 20 mg to normalise BP within 30 minutes

B No antihypertensive; seizures are due to uraemia

C Start enalapril 0.5 mg/kg and furosemide by mouth

D Restrict fluids and recheck BP tomorrow

E IV antihypertensive (labetalol infusion or hydralazine 0.1–0.2 mg/kg) with IV furosemide, lowering BP by no more than 25% of the planned reduction in the first 8 hours

12. A 6-year-old with severe malaria (now on IV artesunate) has passed dark red-brown urine. Dipstick shows blood 3+ but urine microscopy shows no red cells. Hb has fallen from 8 to 5 g/dL. What explains the urine finding?

A PSGN

B Nephrotic syndrome

C Haemoglobinuria from intravascular haemolysis

D Urinary tract infection

E Bladder stone

13. A 7-year-old (22 kg, 120 cm, BSA 0.85 m²) with AKI from sepsis has had 180 mL of urine in the last 24 hours, no other losses, and is clinically euvolaemic. What is the MOST appropriate total fluid intake for the next 24 hours? Advanced

A Full maintenance: about 1,540 mL

B About 520 mL (insensible ≈400 mL/m² ≈ 340 mL + 180 mL urine output), adjusted by daily weight

C Nil by mouth and no IV fluids

D Twice maintenance to force a diuresis

E 180 mL only, to match urine output

14. A 5-year-old with nephrotic syndrome has received prednisolone 60 mg/m²/day for 4 weeks. Dipstick proteinuria remains 3+ daily; she is adherent. What is the BEST interpretation and plan?

A Likely steroid-resistant nephrotic syndrome: refer to paediatric nephrology for confirmation, renal biopsy, genetic testing if available, ACE inhibitor and calcineurin inhibitor therapy

B Frequently relapsing nephrotic syndrome: start levamisole

C Steroid-dependent nephrotic syndrome: taper slowly

D Normal response: continue the same dose for 6 months

E Give albumin infusions weekly until remission

15. Which of the following is an indication for urgent dialysis in a child with AKI?

- **A** Creatinine of 90 µmol/L in a 10-year-old with good urine output
- **B** K^+ 5.6 mmol/L with a normal ECG
- **C** Mild periorbital oedema
- **D** Pulmonary oedema not responding to furosemide in an oliguric child
- **E** Microscopic haematuria

Answers & explanations

1. C · Fluid restriction to insensible losses plus urine output, no added salt, furosemide 1–2 mg/kg, antihypertensive (e.g. nifedipine or amlodipine), and penicillin for 10 days

C is correct: this is **PSGN** (nephritic picture 3–6 weeks after skin infection) with hypertension and AKI; management is fluid and salt restriction, furosemide, BP control and a 10-day course of penicillin to clear the strain. A is wrong: maintenance fluids worsen overload and hypertension. B is wrong: steroids are not indicated in typical PSGN. D is wrong: ACE inhibitors risk hyperkalaemia and a further fall in GFR in acute oliguric GN. E is wrong: a hypertensive oliguric child needs admission and monitoring.

2. E · Prednisolone 60 mg/m²/day (2 mg/kg/day, max 60 mg) once daily for 4 weeks, then 40 mg/m² on alternate days for 4 weeks, with a no-added-salt diet

E is correct: this is **typical idiopathic nephrotic syndrome** (age 1–12, no haematuria, normal BP, creatinine and C3), treated with prednisolone per IPNA 2023/KDIGO. A is wrong: diuretics alone do not induce remission and risk hypovolaemia. B is wrong: biopsy is for atypical or steroid-resistant disease. C is wrong: cyclophosphamide is a steroid-sparing agent for FRNS/SDNS. D is wrong: albumin is reserved for symptomatic oedema or hypovolaemia and does not treat the disease.

3. A · Stop furosemide and give 20% albumin 1 g/kg (or 10–20 mL/kg crystalloid/4.5% albumin if unavailable), then reassess

A is correct: tachycardia, cold peripheries, low BP, oliguria and rising haematocrit after diuretics mean **intravascular hypovolaemia** in a nephrotic child; stop the diuretic and expand the circulation, ideally with albumin. B is wrong: more diuretic worsens shock and AKI and increases thrombosis risk. C is wrong: peritonitis is possible with abdominal pain, but the picture (no fever, soft abdomen, haemoconcentration after diuretics) fits hypovolaemia; antibiotics would be added if peritonitis was suspected, never laparotomy first. D is wrong: 40 mL/kg rapidly risks pulmonary oedema; give measured volumes and reassess. E is wrong: fluid restriction worsens hypovolaemia.

4. D · He has KDIGO stage 3 AKI (creatinine >3× expected; anuria ≥12 h), most likely pre-renal, and needs rehydration followed by reassessment of urine output and creatinine

D is correct: creatinine more than 3 times the age-expected value and anuria for 12 h meet **KDIGO stage 3**, and the history points to **pre-renal AKI** from dehydration, which usually responds to fluids (Schwartz eGFR ≈ 36.5 × 85 ÷ 96 ≈ 32). A is wrong: adult ranges do not apply to toddlers. B is wrong: the rise is over 3 times expected. C is wrong: dialysis is for AEIOU indications not corrected by treating the cause; rehydrate first. E is wrong: diuretics in a dehydrated child worsen hypovolaemia.

5. B · Calcium gluconate 10% 12 mL (0.5 mL/kg) IV over 5–10 minutes with ECG monitoring

B is correct: hyperkalaemia with ECG changes needs **IV calcium first** to stabilise the myocardium (0.5 mL/kg, max 20 mL), followed by salbutamol and insulin–glucose to shift K^+, and removal (resonium or dialysis). A is wrong: resins act over hours. C is wrong: furosemide is slow and unreliable in oliguric AKI. D is wrong: bicarbonate is not first line and only helps if acidotic. E is wrong: salbutamol is right as the second step, but waiting 6 hours is dangerous.

6. C · Refer to paediatric nephrology for evaluation including renal biopsy (possible MPGN, C3 glomerulopathy or lupus)

C is correct: in PSGN, **C3 normalises within 6–8 weeks**; persistence beyond 8 weeks suggests another diagnosis (MPGN, C3 glomerulopathy, lupus) needing biopsy. A is wrong: microscopic haematuria can persist up to a year, but a low C3 should not. B is wrong: steroids should not be started without a histological diagnosis. D is wrong: antibiotics do not affect complement. E is wrong: immunosuppression needs a diagnosis first.

7. A · Delay the live vaccine until at least 1 month after stopping high-dose steroids

A is correct: after **high-dose steroids** (≥2 mg/kg/day or ≥20 mg/day for ≥14 days), live vaccines are deferred until **1 month after stopping**. She is still on alternate-day steroids. B is wrong: remission does not remove the immunosuppressive effect of current steroids. C is wrong: live vaccines are given once off high-dose steroids. D is wrong: albumin level is not the criterion. E is wrong: immunoglobulin is for post-exposure situations, not routine immunisation.

8. E · Prednisolone 2 mg/kg/day (32 mg; or 60 mg/m²) daily until protein-free for 3 days, then 1.5 mg/kg (24 mg; or 40 mg/m²) on alternate days for 4 weeks

E is correct: an infrequent **relapse** is treated with daily prednisolone until remission for 3 days, then alternate-day for 4 weeks (KDIGO/IPNA). A is wrong: relapses get a shorter course than the first episode. B is wrong: steroid-sparing agents are for FRNS/SDNS. C is wrong: a confirmed relapse (3+ for 3 days) is treated, even without oedema, to prevent complications (some clinicians wait a few days if triggered by a cold, but treatment is standard). D is wrong: biopsy is not needed for a steroid-sensitive relapse.

9. D · Spontaneous bacterial peritonitis (usually pneumococcal); blood culture (± ascitic tap) and IV ceftriaxone

D is correct: fever, abdominal pain and peritonism in a nephrotic child on steroids = **primary (spontaneous) bacterial peritonitis**, most often pneumococcal; culture and give IV ceftriaxone urgently (steroids can mask signs). A is wrong: gastritis does not cause fever and peritonism. B is wrong: hypovolaemia causes pain but not fever and guarding with good perfusion. C is wrong: surgery without antibiotics delays treatment; SBP is far more likely. E is wrong: renal vein thrombosis causes flank pain and haematuria, not peritonitis with fever.

10. B · Refer to paediatric nephrology for renal biopsy and specialist care (atypical nephrotic syndrome, possible HIVAN/FSGS)

B is correct: she has **atypical** features (age ≥12, haematuria, hypertension, raised creatinine, HIV), so biopsy and specialist care come before steroids; HIVAN/FSGS is likely and often steroid-resistant. A is wrong: blind standard steroids are inappropriate in atypical disease. C is wrong: ART is the key treatment for HIVAN and must continue (with renal dose adjustment). D is wrong: symptom control alone misses the diagnosis. E is wrong: immunosuppression needs a histological diagnosis.

11. E · IV antihypertensive (labetalol infusion or hydralazine 0.1–0.2 mg/kg) with IV furosemide, lowering BP by no more than 25% of the planned reduction in the first 8 hours

E is correct: this is **hypertensive encephalopathy** in PSGN; lower BP in a controlled way (≤25% of the planned reduction in 8 h, then over 24–48 h) using titratable IV agents plus furosemide for volume overload. A is wrong: a large, rapid fall risks cerebral ischaemia and blindness, and sublingual nifedipine is unpredictable. B is wrong: the history and BP point to hypertensive encephalopathy, not uraemia. C is wrong: ACE inhibitors risk hyperkalaemia and are too slow. D is wrong: delay risks further seizures and brain injury.

12. C · Haemoglobinuria from intravascular haemolysis

C is correct: dipstick blood without red cells on microscopy means **free haemoglobin (haemoglobinuria)** from intravascular haemolysis (malaria/'blackwater', G6PD), a cause of AKI; monitor urine output and creatinine and transfuse as indicated. A is wrong: PSGN shows dysmorphic RBCs and casts. B is wrong: nephrotic syndrome gives heavy proteinuria, not haemoglobinuria. D is wrong: UTI gives leucocytes and nitrites. E is wrong: stones cause RBCs in the urine.

13. B · About 520 mL (insensible ≈400 mL/m² ≈ 340 mL + 180 mL urine output), adjusted by daily weight

B is correct: in oliguric, euvolaemic AKI give **insensible losses (~400 mL/m²/day) plus urine output** (≈340 + 180 mL), and use daily weight to fine-tune. A is wrong: full maintenance (Holliday–Segar ≈1,540 mL) in oliguria causes overload. C is wrong: withholding all fluid causes dehydration and hypoglycaemia. D is wrong: forced diuresis by fluid loading causes pulmonary oedema. E is wrong: matching urine alone ignores insensible losses.

14. A · Likely steroid-resistant nephrotic syndrome: refer to paediatric nephrology for confirmation, renal biopsy, genetic testing if available, ACE inhibitor and calcineurin inhibitor therapy

A is correct: **no remission after 4 weeks** of full-dose prednisolone defines (likely) **SRNS**; IPNA advises a short confirmation period and specialist work-up with biopsy, genetics, RAS blockade and a calcineurin inhibitor. SRNS is more common in African children. B is wrong: FRNS describes relapses after remission. C is wrong: SDNS also requires an initial response. D is wrong: prolonged high-dose steroids add toxicity without benefit. E is wrong: albumin does not induce remission.

15. D · Pulmonary oedema not responding to furosemide in an oliguric child

D is correct: **fluid overload with pulmonary oedema refractory to diuretics** is a dialysis indication (the 'O' in AEIOU). A is wrong: a mild creatinine rise with good output is managed medically. B is wrong: mild hyperkalaemia without ECG changes is treated medically. C is wrong: mild oedema is managed with salt/fluid restriction and diuretics. E is wrong: haematuria alone does not need dialysis.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Cola urine after scabies

Waiswa, a 10-year-old boy from Iganga, is brought to the paediatric ward with swelling of his face for 4 days and 'urine like Coca-Cola'. His mother says he is passing very little urine and complains of headache.

1. What history will you take?

> **Sore throat 1–2 weeks** or **skin sores/scabies 3–6 weeks** ago (and family members with scabies), urine colour and volume, swelling pattern, **headache, vomiting, blurred vision, fits**, breathlessness/orthopnoea, rash or joint pains (HSP, SLE), drugs/herbs (NSAIDs), previous kidney problems, HIV status, immunisation, nutrition.

2. History: infected scabies 3 weeks ago. Exam: 28 kg, 132 cm, periorbital and ankle oedema, BP 146/98 (>99th centile), HR 92, RR 24, no crackles, liver 1 cm, healed scabies lesions. Dipstick blood 3+, protein 1+. What is your working diagnosis and what tests do you order?

> **PSGN** with hypertension and probable AKI. Tests: **creatinine/urea, Na, K**, **C3 (± C4)**, **anti-DNase B** (skin) and ASO, urine microscopy (RBC casts), FBC, malaria RDT, CXR if breathless, renal ultrasound if atypical.

3. Results: creatinine 120 µmol/L, K^+ 5.4, Na 133, C3 low, anti-DNase B high. Urine output 0.5 mL/kg/h over the last 8 h. Write the management plan.

> Admit; **daily weight, strict input/output, BP 4-hourly**. **Fluid**: BSA ≈ √(132×28/3600) ≈ 1.01 m² → insensible ≈ 400 mL/day + yesterday's urine output. **No added salt; low-K^+ foods**. **Furosemide 1–2 mg/kg** (28–56 mg) oral/IV. **Amlodipine 0.1 mg/kg/day** (≈2.5–5 mg) ± nifedipine 0.25 mg/kg as needed. **Penicillin V 250 mg 6-hourly × 10 days** or BPG 0.6 MU once (<30 kg). **Treat scabies** (permethrin; family too). No steroids. Daily creatinine and K^+.

4. On day 2 he has a severe headache, vomits and has a seizure. BP 172/112. What do you do?

> **Hypertensive encephalopathy**: ABC, recovery position, **diazepam 0.3 mg/kg IV** (or 0.5 mg/kg rectal) if still fitting; check glucose and Na. **IV hydralazine 0.1–0.2 mg/kg** (≈3–5 mg) or **labetalol infusion**, plus **IV furosemide 2 mg/kg**. Reduce BP **by no more than 25% of the planned fall in the first 8 h**, then gradually. Call a senior; monitor neuro obs.

5. When will you worry this is not simple PSGN?

> If **creatinine keeps rising** or oliguria persists >1 week (RPGN/crescents), **nephrotic-range proteinuria** persists, **C3 low beyond 8 weeks**, hypertension >6 weeks, gross haematuria >3 weeks, or systemic features (rash, arthritis). Then refer for **biopsy at Mulago**.

6. What follow-up and prognosis do you explain to the family?

> Most children recover fully (>95%). Swelling and BP settle in 1–3 weeks; **C3 normal by 8 weeks**; microscopic blood may last up to a year. Follow-up: BP and urine at 2 and 6 weeks, **C3 at 8 weeks**, then 3, 6 and 12 months. Prevent recurrence of skin infections and scabies at home.

Take-away: PSGN is an overfilled kidney: restrict fluid and salt, give furosemide and control BP carefully, treat the strep and scabies, and follow C3 to 8 weeks.

Case 2: The swollen 3-year-old

Nalwoga, a 3-year-old girl, is brought to Mengo with 2 weeks of swelling that started around her eyes and has spread to her legs and tummy. Her grandmother thought it was 'too much salt'.

1. What examination and bedside tests do you do first?

> Weight (compare with the child health card), height, **BP with correct cuff**, HR, **CRT and peripheral temperature**, oedema map (eyes, sacrum, legs, labia, ascites, effusions), abdominal tenderness, skin infections, signs of malnutrition (kwashiorkor). **Urine dipstick**: protein 4+, blood negative.

2. Findings: 15 kg (card weight 13 kg), height 94 cm, BP 92/56, warm, CRT <2 s, ascites, no tenderness. What bloods and screening do you send?

> **Albumin** (12 g/L), **cholesterol** (high), **creatinine, Na, K**, FBC, **C3**, **UPCR** if available; **HIV, HBsAg, syphilis**; **TB screen** (contact history, CXR); malaria RDT; stool for parasites. Deworm with albendazole; praziquantel if schistosomiasis risk.

3. All screening normal, C3 normal, creatinine 24 µmol/L. Write the prednisolone prescription (use pre-oedema weight 13 kg; BSA ≈ 0.58 m²).

> **Prednisolone 60 mg/m²/day ≈ 35 mg** (2 mg/kg ≈ 26 mg; max 60) **once each morning for 4 weeks**, then **40 mg/m² ≈ 23 mg (round to 25 mg) on alternate days for 4 weeks**, then stop (IPNA 2023/KDIGO). Give with food. (UCG and some Ugandan units use 6 weeks daily + taper/6 weeks alternate-day; follow your unit.)

4. Her oedema is moderate and she is comfortable. Will you give furosemide or albumin?

> **No routine diuretics or albumin.** No-added-salt diet; normal fluids unless hyponatraemic. Albumin 20% 0.5–1 g/kg with furosemide only for **symptomatic oedema** (breathing difficulty, skin breakdown) or **hypovolaemia**. Diuretics alone risk hypovolaemia and thrombosis.

5. On day 8 she develops fever 39 °C, abdominal pain and guarding. What is your concern and action?

> **Spontaneous bacterial peritonitis** (pneumococcus) while on steroids. Blood culture (± diagnostic ascitic tap), **IV ceftriaxone 80–100 mg/kg/day**, assess perfusion (hypovolaemia also causes abdominal pain), surgical review if unclear.

6. She enters remission on day 10. What do you teach the family before discharge?

> **Home dipstick** of first-morning urine (daily during illness/relapse, otherwise weekly), record in a diary; **relapse = 3+ for 3 days** → return; finish the steroid course exactly; **steroid card**; signs of infection, abdominal pain, reduced urine or limb swelling → come immediately; **no live vaccines** until 1 month after high-dose steroids; PCV/PPSV23 and influenza; no added salt; relapses are common (70–80%) and usually respond again.

Take-away: Typical childhood nephrotic syndrome is steroid-sensitive. Screen for secondary causes and TB, give weight/BSA-based prednisolone (dose on dry weight), avoid unnecessary diuretics, watch for infection and hypovolaemia, and teach home monitoring.

Case 3: No urine after diarrhoea and malaria

Kyomuhendo, a 2-year-old boy, is referred from a HC III in Hoima with 4 days of diarrhoea and fever, now drowsy. He has not passed urine since the morning (14 hours).

1. What are your first actions?

> ETAT: ABC, SpO_2, **glucose**, conscious level. Assess **dehydration/shock** (CRT, pulse, cold hands, sunken eyes, skin pinch). IV access; bloods for **creatinine, Na, K, Hb, malaria RDT, glucose**, blood culture. **ECG/monitor** if available (hyperkalaemia).

2. Findings: 11 kg, CRT 4 s, weak fast pulse 170, sunken eyes, drowsy. RDT positive. Glucose 4.1. Hb 7.8. What fluids do you give?

> **Severe dehydration with shock** (not SAM, Hb >5): start **WHO Plan C**: **Ringer's lactate 30 mL/kg (330 mL) over 30 minutes** (age ≥12 months; repeat once if the radial pulse is still weak), then **70 mL/kg (770 mL) over 2.5 hours**, reassessing every 15–30 minutes and starting ORS as soon as he can drink. **IV artesunate 3 mg/kg** for severe malaria. Ceftriaxone if sepsis possible.

3. Results: creatinine 160 µmol/L, urea 22 mmol/L, K^+ 6.6, Na 129, HCO_3^- 12. ECG: peaked T waves. What now?

> **KDIGO stage 3 AKI** with **dangerous hyperkalaemia**. **Calcium gluconate 10% 5.5 mL (0.5 mL/kg) IV over 5–10 min** with monitoring; **salbutamol 2.5 mg neb**; **insulin 1.1 units + glucose 5.5 g (55 mL of 10% dextrose)** IV over 30 min with hourly glucose; **sodium bicarbonate 1–2 mmol/kg** for acidosis; remove K^+ from fluids; resonium 1 g/kg. Catheterise and measure urine output.

4. After rehydration he is perfused but passes only 20 mL in 6 hours. What is your fluid plan and what else must you do?

> Now **euvolaemic and oliguric** (likely ATN from shock/malaria): restrict to **insensible (~400 mL/m²/day; BSA ≈0.5 m² → ~200 mL/day) + urine output + losses**, adjusted by weight. Trial **furosemide 1–2 mg/kg IV** once. Stop nephrotoxins (no NSAIDs; avoid gentamicin). Adjust drug doses. Repeat K^+ and creatinine 6–12-hourly. Nutrition with adequate calories, low K^+.

5. When will you refer for dialysis?

> **AEIOU**: K^+ ≥7 or rising despite treatment, refractory acidosis, pulmonary oedema/fluid overload >10%, uraemic encephalopathy or bleeding, **anuria >24 h** or rising creatinine. Refer early to **Mulago paediatric nephrology** for **peritoneal dialysis** before he deteriorates.

6. How do you present him?

> "Kyomuhendo, 2 years, 11 kg (WHZ –1.1), immunised per UNEPI, breastfed plus family foods, HIV-unexposed, admitted with severe malaria and acute watery diarrhoea causing hypovolaemic shock and KDIGO stage 3 AKI with hyperkalaemia (K^+ 6.6, ECG changes) and metabolic acidosis. Resuscitated with 20 mL/kg Ringer's, IV artesunate, and treated for hyperkalaemia. Now euvolaemic but oliguric; on restricted fluids with 6-hourly electrolytes; referral for PD if AEIOU criteria develop."

Take-away: In Ugandan children, AKI usually starts as pre-renal (diarrhoea, malaria, sepsis). Restore perfusion fast, then restrict fluids once oliguric and euvolaemic, treat potassium aggressively, avoid nephrotoxins, and refer early for peritoneal dialysis.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.